

Live Meeting Summarizer Application

Module 1: Speech-to-Text System Setup

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Project Overview



Objective

Build an end-to-end system that captures meeting audio in real-time, converts speech to text using deep learning models, and generates concise AI-powered summaries.

Module 1 Scope

- Speech-to-Text conversion pipeline
- Model selection and comparison
- Dataset preparation and evaluation
- Word Error Rate analysis

System Architecture Pipeline

Meeting Audio (.wav)



Audio Processing (FFmpeg)



Whisper Speech-to-Text Model



Generated Transcript



WER Evaluation (jiwer)

Dataset: AMI Meeting Corpus



Meeting Recordings

171 hours of multi-speaker meeting audio with microphone array recordings



Transcripts

Aligned text transcripts with speaker labels and timing information



Speaker Diversity

100+ speakers across 100+ meetings with natural conversation patterns



Realistic Scenarios

Business meetings, discussions, and collaborative sessions

The AMI corpus provides realistic meeting scenarios with challenging acoustics, overlapping speech, and diverse speaking styles—making it ideal for evaluating meeting transcription systems.

Model Comparison: Whisper vs. Vosk

Feature	Whisper	Vosk
Accuracy	State-of-the-art WER	Moderate accuracy
Language Support	99+ languages	Limited languages
Architecture	Transformer-based	HMM + DNN
Offline Capability	Yes (downloaded)	Yes (lightweight)
Training Data	680K hours	Limited datasets
Implementation	PyTorch + OpenAI	Kaldi-based

Decision: Whisper selected for superior accuracy and multilingual support despite larger model size.

Implementation Setup

01

Python Environment

Python 3.9+ with virtual environment management

02

PyTorch Framework

Torch 1.13+ for deep learning model loading and inference

03

Whisper Model

OpenAI's Whisper base model for highest accuracy transcription

04

FFmpeg Processing

Audio preprocessing and format conversion utilities

05

jiwer Evaluation

Word Error Rate calculation and analysis toolkit

Word Error Rate (WER)

Evaluation Formula

$$WER = \frac{S + D + I}{N} \times 100\%$$

Where **S** = substitutions, **D** = deletions, **I** = insertions, and **N** = total reference words

Example Results

- Whisper-large: 8.2% WER
- Whisper-medium: 12.5% WER
- Vosk: 18.7% WER
- Baseline: 25.3% WER



Interpretation

Lower WER indicates better transcription quality. Industry standard for meetings: <10% WER is excellent, 10-15% is acceptable, >15% requires improvement.

Module 1 Achievements & Next Steps

1

Complete Pipeline

Successfully implemented end-to-end speech-to-text system with Whisper integration

2

Dataset Integration

Processed AMI corpus and validated data quality for training and evaluation

3

Model Evaluation

Established WER benchmark and demonstrated Whisper's superior performance

4

Technical Foundation

Built reusable codebase and documentation for future modules

Module 2: Coming Next

Speaker Diarization (who spoke when) and AI-Powered Meeting Summarization using transformer-based language models